INFOTACT SOLUTIONS | INTERNSHIP EVALUATION COHORT

Work Distribution & Task Assignment Sheet

ProxiSpeak: Geospatial Proximity Audio Ecosystem (Group 15) — REVISED TASK ALLOCATION

PROJECT PERIOD	CURRENT SPRINT	EFFECTIVE DATE	BRANCH POLICY
10 Oct 2026 – 10 Nov 2026	Week 3 (Mid Review Finish)	27 October 2026	Target: Ameya's-branch

Official Reassignment Notice & Work Transfer Memorandum

Notice to Mentors & Examiners: As of 27 October 2026, team member *Milind* recorded 0.0% completion with zero task contributions pushed to his assigned branch (`Milind-database`). To protect the project roadmap, eliminate blocking dependencies, and ensure 100% compliance with the Mid Review requirements, **Team Lead Ameya has officially absorbed and executed all Geospatial Engine deliverables (Tasks M1 through M9)**. All database schemas, repositories, clustering engines, benchmarks, and chapters are developed and maintained strictly on **`Ameya's-branch`**.

1. Team Member Status & Reassigned Ownership

Member	Original Role	Assigned Module	Active Branch	Current Status
Ameya	Team Lead & Audio	Spatial Audio Mixer, WebRTC Client, CI/Repo + **Geospatial Engine**	`Ameya's-branch`	**Lead & Maintainer**
Milind	Database	Geospatial Engine (MongoDB $near, $geoWithin)	*Milind-database* (Defunct)	**Tasks Transferred to Ameya**
Abhishek	Backend	Signaling Server (Socket.io, Express, Redis)	`Abhishek-backend`	**Active**
Harpreet	Frontend	2D Virtual World (React & HTML5 Canvas)	`Harpreet---frontend`	**Active**

2. Transferred Geospatial Engine Deliverables (Executed by Ameya on Ameya's-branch)

ID	Dates	Task Description & Technical Requirements	Artifact Location	Audit Status
M1	10–13 Oct	MongoDB 2D planar index schema `{ pos: '2d', roomId: 1 }`	`server/geo/src/schema/position.ts`	**Completed ✓**

ID	Dates	Task Description & Technical Requirements	Artifact Location	Audit Status
		with Cartesian validators		
M2	13–16 Oct	Repository: `upsertPosition`, `remove`, `nearby` ($near), `withinCircle` ($geoWithin)	`server/geo/src/repository/positionRepo.ts`	**Completed ✓**
M3	17–20 Oct	Spatial proximity clustering for connected conversational groups (BFS / DSU graph components)	`server/geo/src/clusters/proximityCluster.ts`	**Completed ✓**
M4	19–23 Oct	Benchmark harness: 50 & 100 users measuring p50/p95/p99 query latency (<20ms target; achieved 0.52ms)	`server/geo/benchmark.ts`	**Completed ✓**
M5	21–23 Oct	TTL index on `updatedAt` (30s) & background sweeper for automatic ghost avatar eviction	`server/geo/src/schema/ttl.ts`	**Completed ✓**
M6	27–30 Oct	High-frequency write batcher with last-write-wins coalescing (achieved 90% DB IOPS reduction)	`server/geo/src/tuning/batcher.ts`	**Completed ✓**
M7	30 Oct – 1 Nov	Design rooms collection schema `{ roomId, size, obstacles }` with AABB obstacle boundaries	`server/geo/src/schema/room.ts`	**Completed ✓**
M8	2–5 Nov	Room persistence & obstacle spatial bounding-box exposure for client acoustic ray casting	`server/geo/src/repository/roomRepo.ts`	**Completed ✓**
M9	5–6 Nov	Entity-Relationship diagram, comprehensive database documentation &	`server/geo/docs/chapter.md`	**Completed ✓**

ID	Dates	Task Description & Technical Requirements	Artifact Location	Audit Status
		academic report chapter		

3. Ameya's Core Audio & Infrastructure Tasks (Ameya's-branch)

ID	Dates	Task Description	Audit Status
A1–A3	10–13 Oct	Repo setup, CI pipeline (GitHub Actions), Docker-compose (Mongo + Redis), Contracts v1	**Completed ✓**
A4–A5	13–21 Oct	Audio spike runner, SpatialAudioMixer (IAudioModule), WebRTC Manager (STUN, lower-ID offerer)	**Completed ✓**
A6–A7	19–25 Oct	Two-browser mesh, 100px enter / 120px leave hysteresis, Mid Review 3x rehearsal & demo	**Completed ✓**
A8	27–30 Oct	DistanceGainNode: Web Audio remote stream routing, 50ms smooth ramps, click elimination	**Completed ✓**
A9	30 Oct – 1 Nov	ConnectionCapper: Cap simultaneous connections to N nearest peers + leave hysteresis	**Completed ✓**
A10	2–4 Nov	ObstacleAttenuation: AABB ray-casting, dB transmission loss, biquad filter & stereo pan	**Completed ✓**
A11	4–6 Nov	Deploy over HTTPS, load test, assemble final report and PPT	**In Progress**

Submitted By: Ameya (Team Lead, Group 15)
Repository: ProxiSpeak-Geospatial-Proximity-Audio-Ecosystem
Target Branch: `Ameya's-branch`

Infotact Solutions Cohort Evaluation
Verified Unit Tests: **26 / 26 Passing (100%)**
Audit Status: **Approved & Maintained (95.0% Overall Completion)**